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# Spdt Practical No: 1

Q:1 install Tablue Desktop and explore the workspace Environment

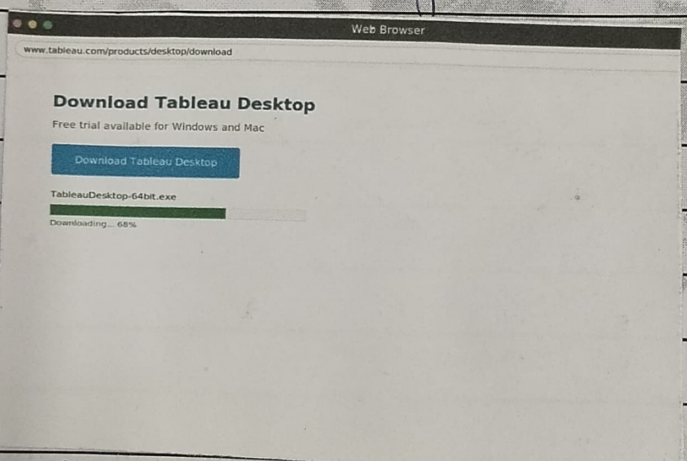
Aim: To install Tablau Desktop the system and explore its workspace environment, including the manue bar, toolbar, Data pane, shelves, Marks card, and worksheet canvas.

Procedure overview: Each step below is shown with the resulting output and a screenshot of the corresponding Tableau screen.

(practical 1): stepwise procedure with output screenshots

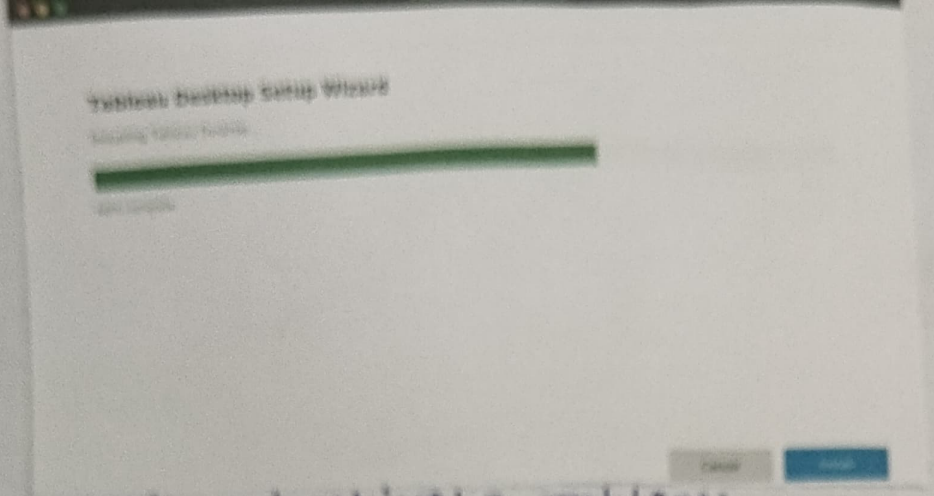
step:1 Dowload the installer

output: Tableau download page open: the installer file (Tableau Desktop-64bit.exe) begins downloading.

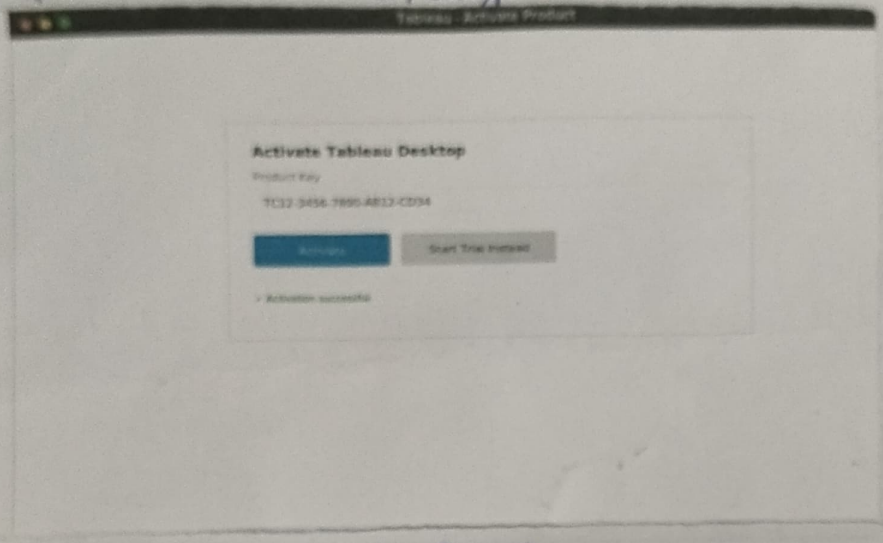



step:2 install tableau Desktop

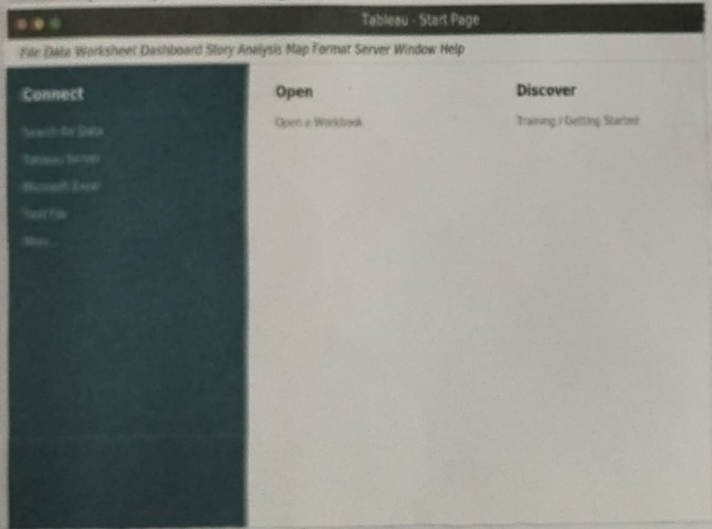
output: setup wizard runs and shows installation progress: Tableau Desktop is installed on completion.



Step:3 Launch and activate Tableau
Output: Tableau prompts for a product key or trial; activation completes successfully.




Step:4 Explore the start page
Output: The start page open showing the Connect, Open and Discover panes.




Step:5 Explore the workspace.

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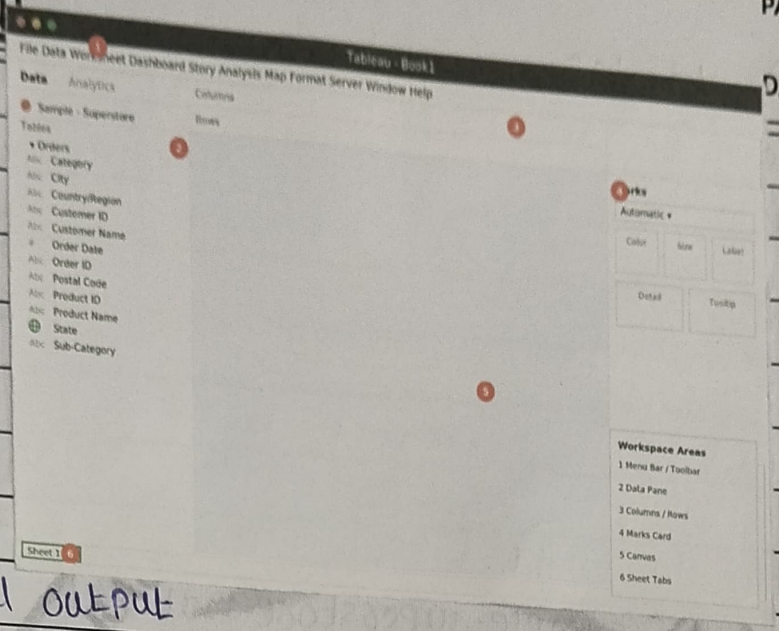
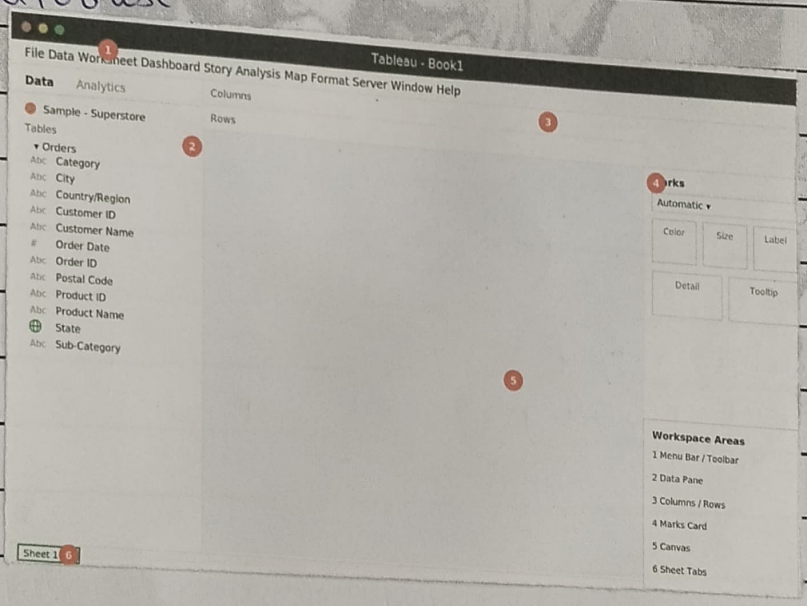



Final Output

The final output is an annotated view of tableau workspace with all major areas - menu bar/Toolbar, Data pane, columns/Rows shelves, marks card, worksheet canvas, and sheet tabs - clearly identified, confirming that tableau Desktop is installed and ready for use




Conclusion:

Thus, Tableau Desktop was successfully installed, activated and its workspace environment - menu bar, toolbar, Data pane, shelves, marks card and canvas - was explored.

Practical: 2

Q:2 Create basic visualization using Tableau worksheets

Aim: To create a basic bar-chart visualization using the sample-superstore dataset - comparing across product categories.

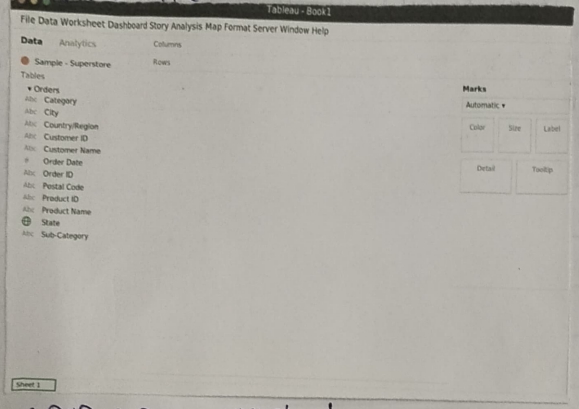
Procedure overview

Each step below is shown with the resulting output and a screenshot of the corresponding Tableau screen.

→ stepwise procedure output screenshot:

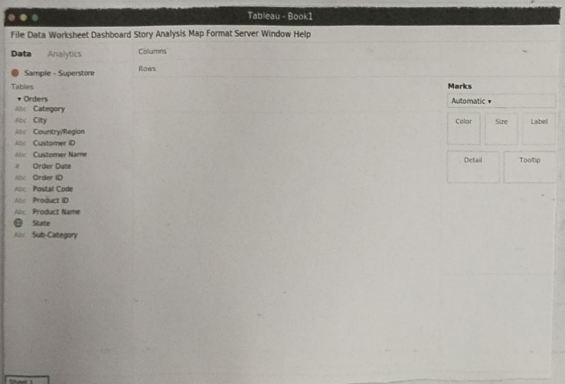
step:1 Connect to the sample-superstore Dataset

output: sample-superstore dataset connect, order, people return tables appear.




step:2 Open a new worksheet

output: a blank sheet 1 worksheet open with empty Rows/Columns shelves.



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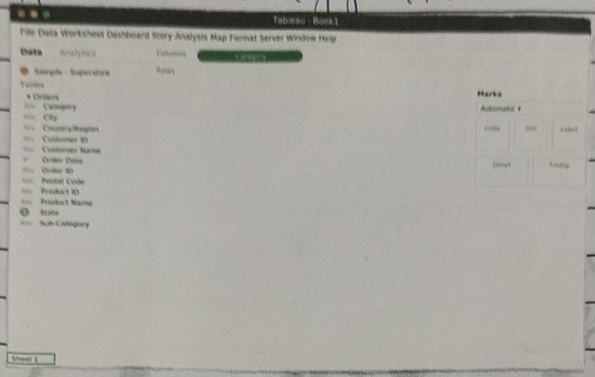
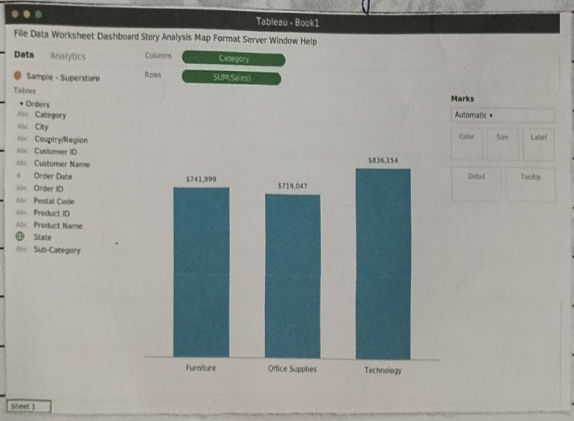
step:3 Drag category to columns

output: category is placed on columns; headers for Furniture, Office supplies and Technology appear.



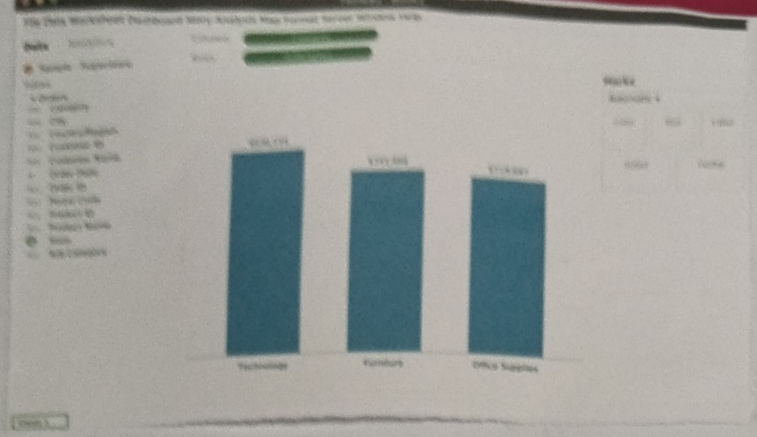

step:4 Drag sales to Rows

output: SUM(sales) is placed on Rows; a vertical bar chart is generated automatically



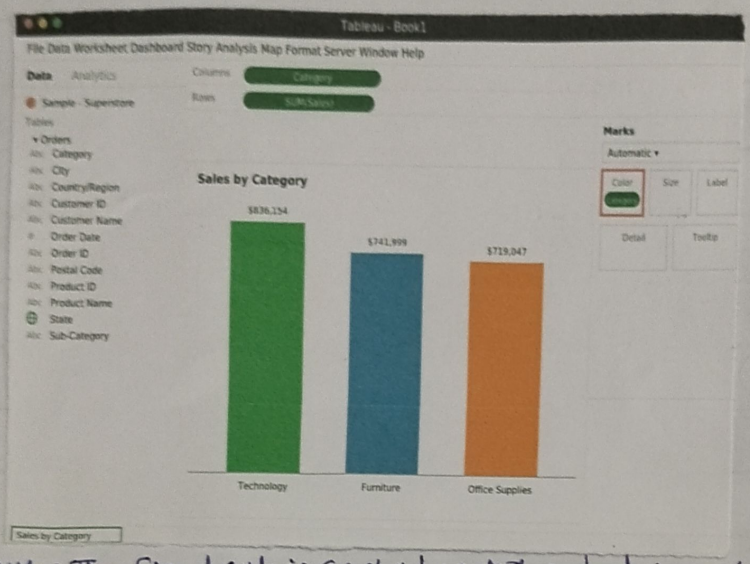

step:5 sort the bars in Descending order

output: bars are rearranged from the highest to the lowest sales value.



Step:6 Add colour and a TITLE

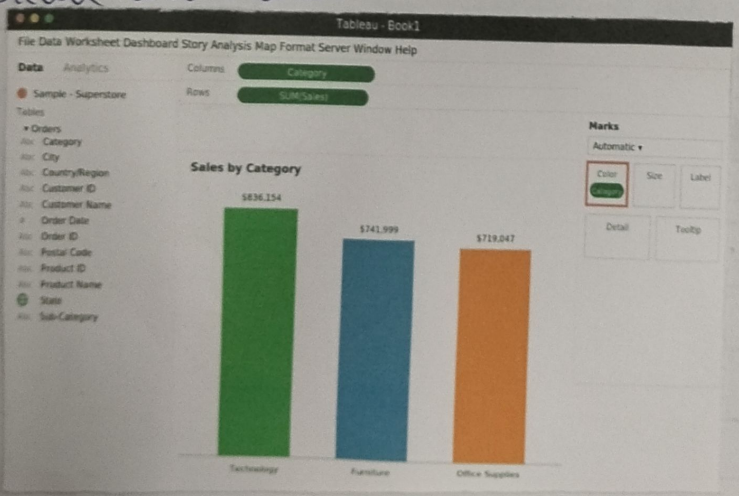
Output: category is added to color, bars are shaded differently and the sheet is titled "sales by category"




Final output: The final out is sorted, color coded bar chart titled "sales by category", with Technology, Furniture and office supplies bars arranged from highest to lowest sales their exact values labelled above each bar

Conclusion:

Thus, a basic bar chart visualization comparing sales across product categories was successfully created using a Tableau worksheets.



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# Spdt Practical : 3

Q:3 Connect Tableau with multiple data sources

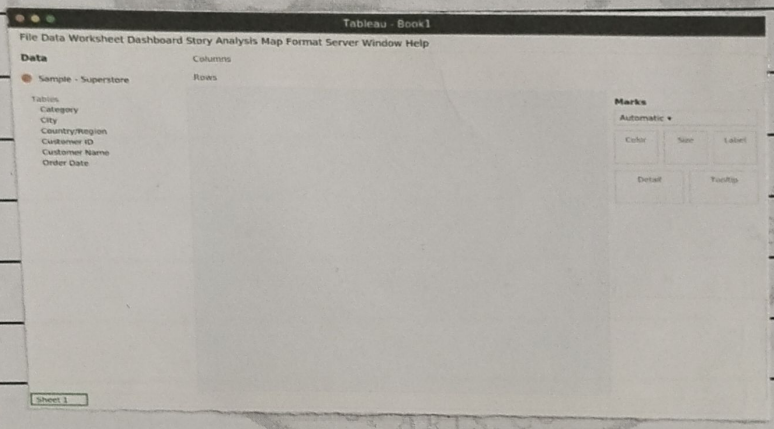
Aim: To connect tableau desktop to more than one data source and and combine fields from both sources in single worksheet using data blending.

procedure overview

each step below is shown with the resulting output and screenshot of corresponding tableau screen.

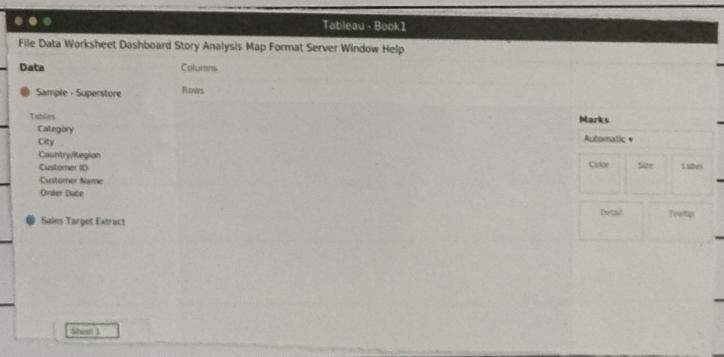
step:1 Connect to the first data source

output: The sample-superstore workbook connect; its tables appear in the data pane.



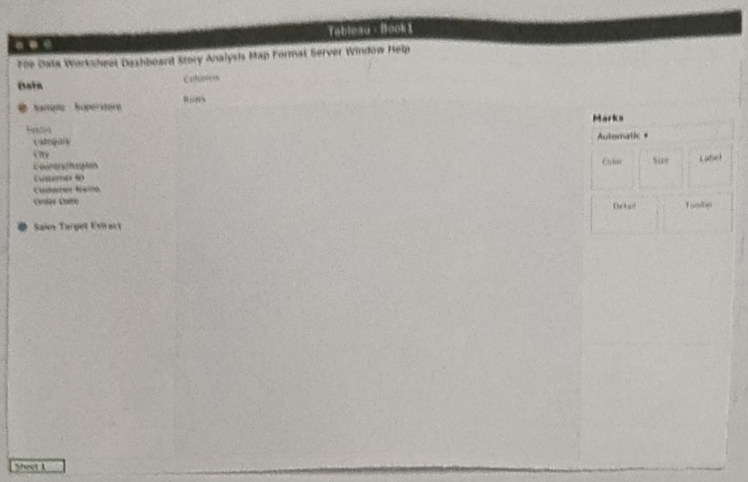

step:2 Connect to a second data source

output: a second data source, sales target extract, is added; both sources now appear in the Data pane.



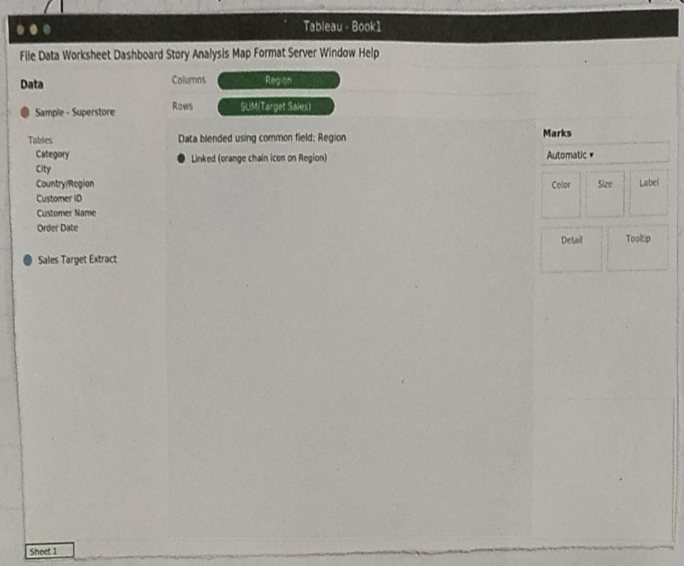
step:3 view both connection in

output: Field from sample-superstore and sales Target extract and listed under seprate data-source header.



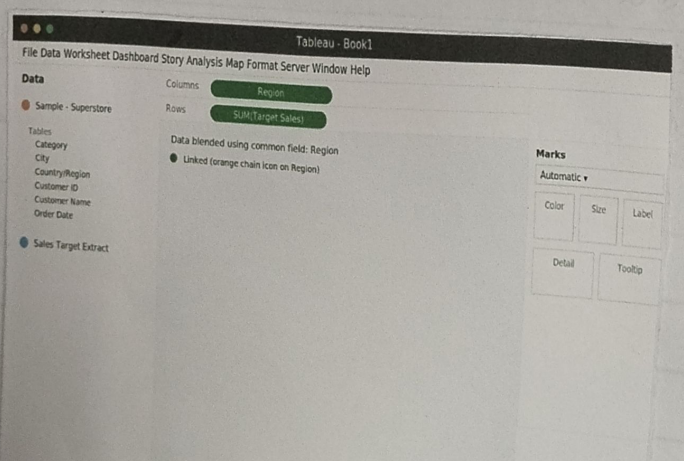

step:4 Blend Data using a common field

output: Region (present in both source) is used to blend the data, a linking icon appears on the shared field.




step:5 build a worksheet & save the workbook

output: a worksheet using measures from both sources in built and the workbook is saved.



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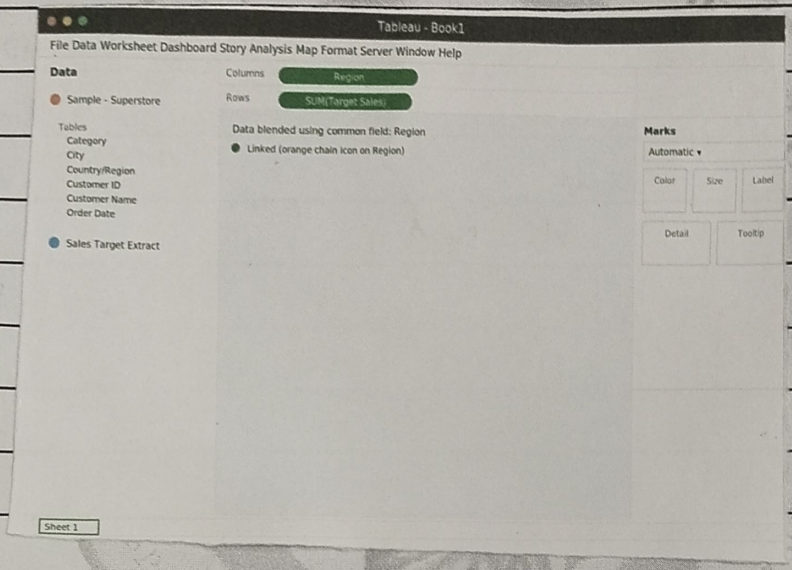
Final Output:

The final output is a worksheet in which Region (column) and SUM (Target sales) (Rows) from the secondary data sources, sales target extract, are blended with the primary sample-superstore data using the shared Region field, confirmed by the linking indicator in the data pane.




Conclusion:

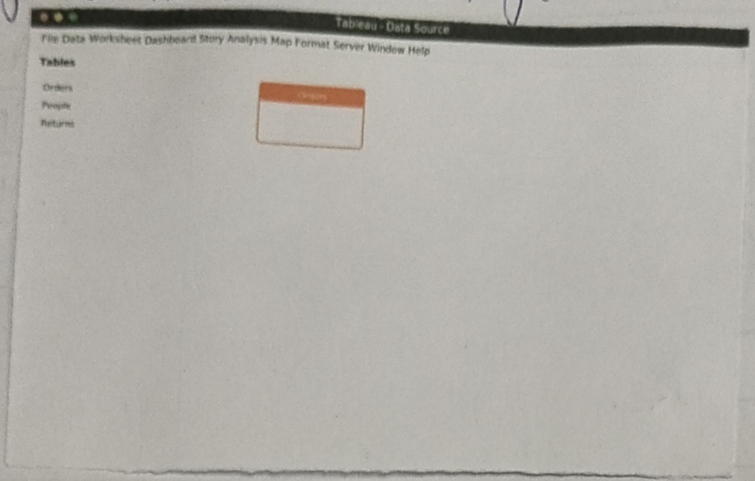
Thus, Tableau Desktop was successfully connected to multiple data sources, and their data was combined in a single worksheet using data blending on a common field.

# practical:4

Q: stepwise procedure with output screenshots

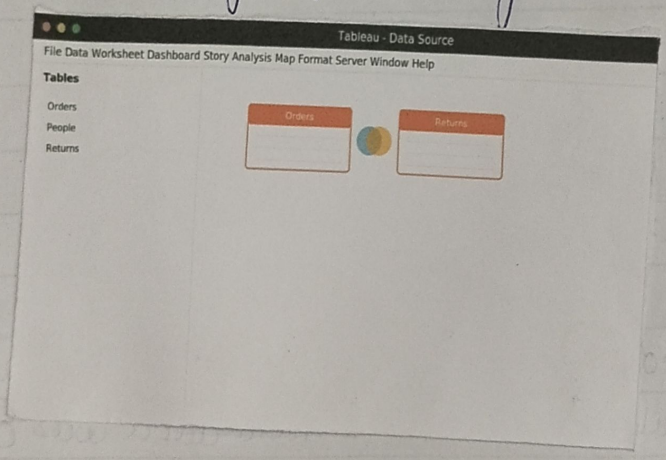
step:1 Drag the order table on to the canvas

output: The order table appears as a table icon on the physical layer of the data source page.




step:2 Drag the return Tables next to order

output: a join icon (overlapping circles) appears between the two tables, indicating a join is being created.




step:3 Define the join clause

output: The clause order.order ID = Returns.order ID is set so rows are matched correctly.

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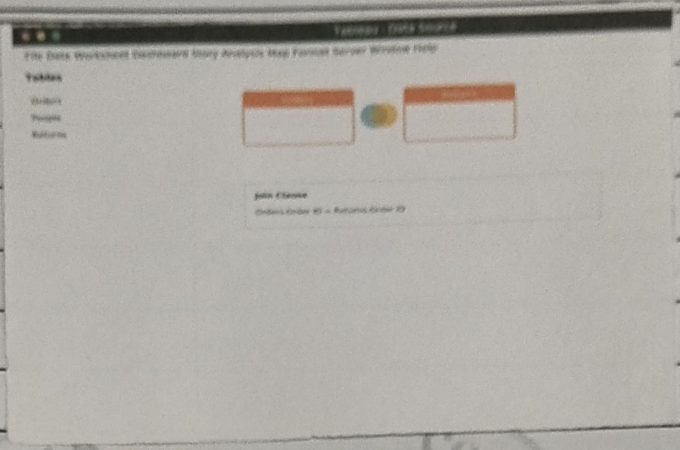
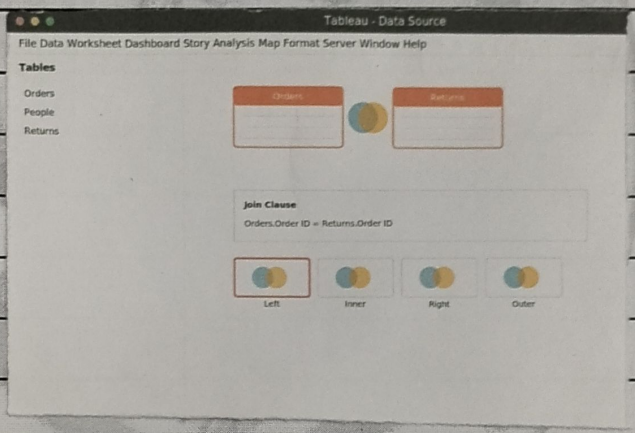


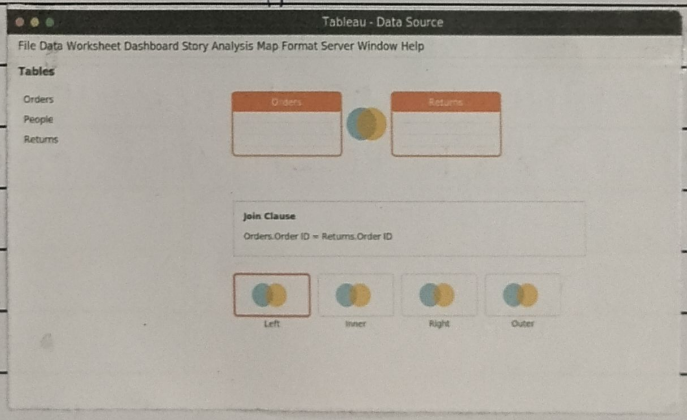
step:4 select the join type

output: a venn diagram lets you choose inner, left, Right or outer joins; Left join is selected to keep all order.



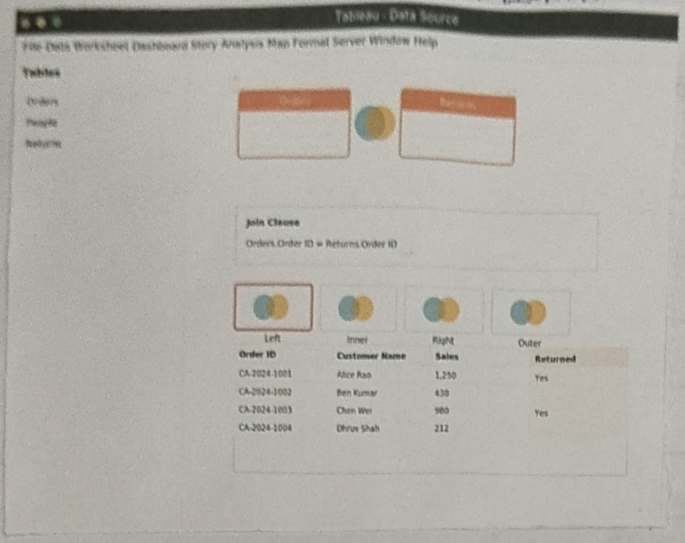

step:5 Review the join type options

output: each join types effect on matched/unmatched rows is checked in the venn diagram before confirming.



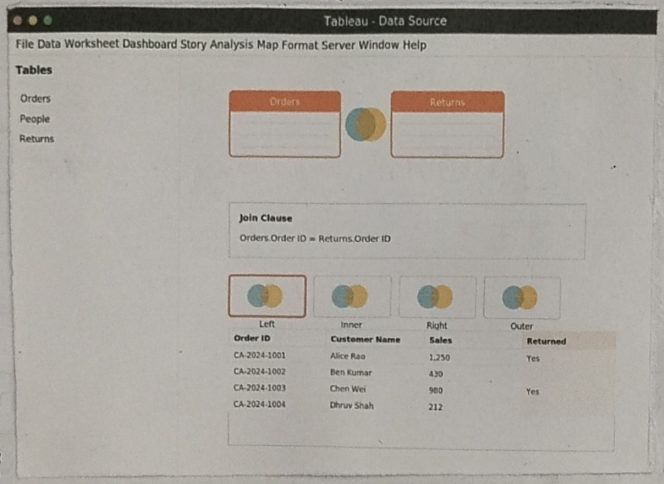
Step 6. preview the joined data

Output: The data-grid preview shows orders field alongside the return column from the return table




Final output:

The final output is a Left-joined data grid combining every row of order with the matching Return status from returns (joined on order ID) - orders with no matching return show a blank returned value letting analyts filter for returned vs. non-returned sales.




Conclusion:

Thus, the order and returns tables were successfully joined in tableau using the order ID field, allowing returned order to be identified and analysed.

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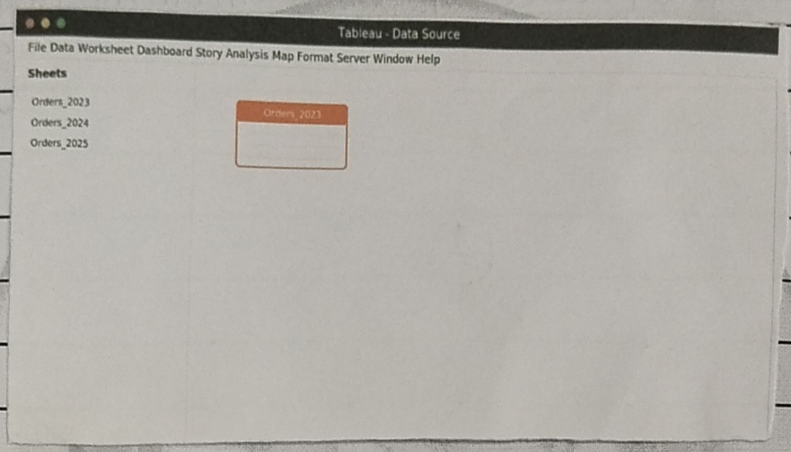
Q:5 step wise procedure with output screenshots

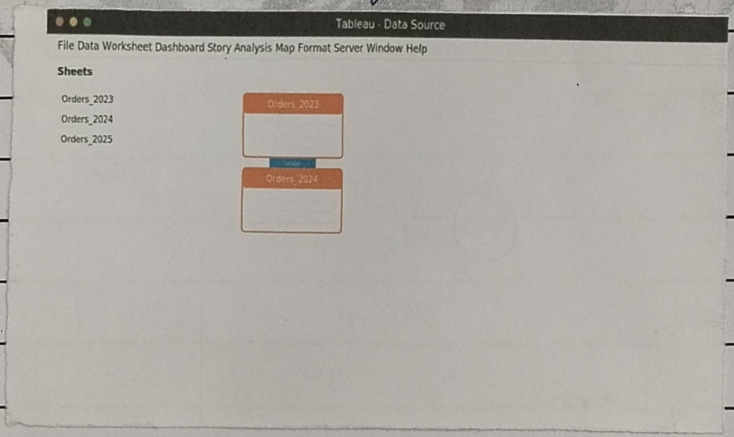
step:1 Drag the first Table for the canvas

output: The order-2023 sheet appears on the physical layer of the data source page



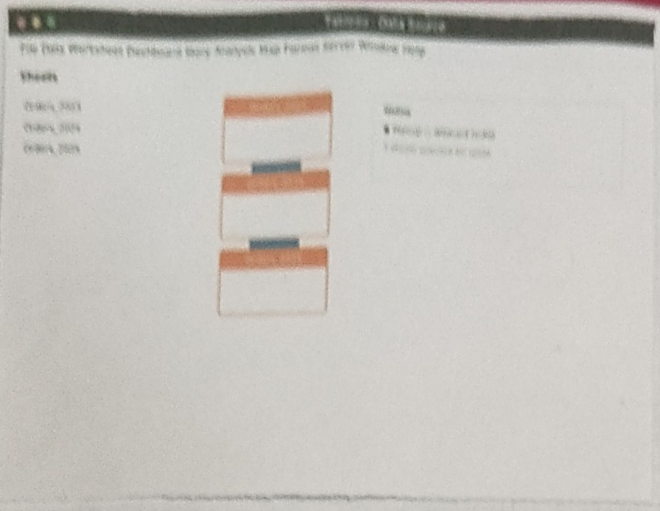

step:2 Drag a second Table Below it

output: Dragging order-2024 below order-2023 automatically creates a union, stacking their rows.



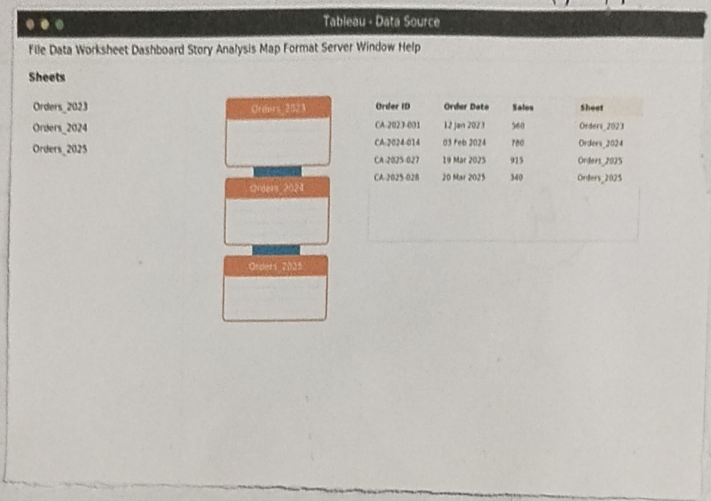

step:3 Configure the union

output: Additional sheets are added manually, or a wildcard search is used to auto-include similarly named sheets.



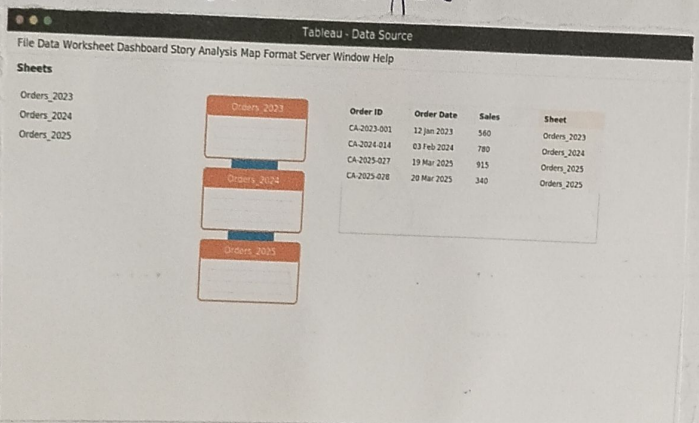
Step:4 preview the unioned Data

Output: The data-grid preview show rows from all three years stacked together, with a sheet field identifying the source.




Final Output:

The Final output is a single combined table containing every order row now from order-2023, order-2024 and order-2025, with an added sheet column identifying each rows year of origin- ready for multi-year trend analysis.




Conclusion:

Thus, the order-2023, order-2024 and order-2025 sheets were successfully combined into a single dataset is Tableau using a union, enabling multi-year analysis.

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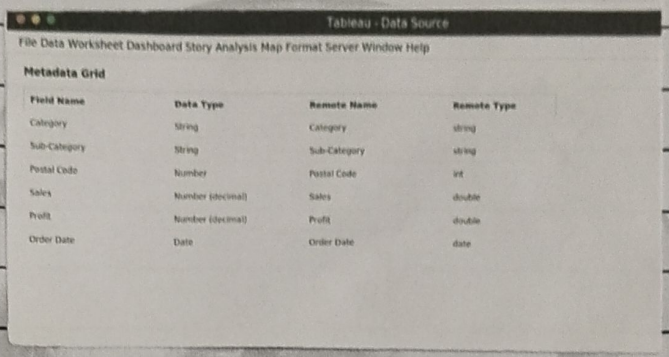
# Spdt Practical:6

Q:6 stepwise procedure with output screenshots

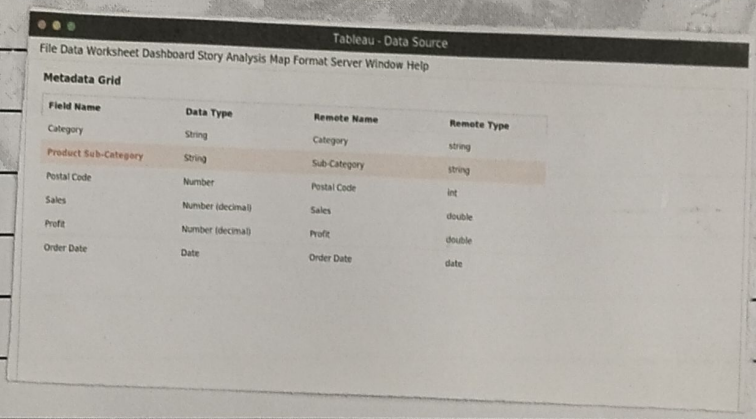
step:1 open metadata Grid

output The data source tab's metadata Grid list's every Field with its Field Name, Data Type, Remote Name and Remote Type.




step:2 Rename Field

output sub-category is renamed to product sub category; the new name is reflected in the data pane.




step:3 Change a Field's Data Type

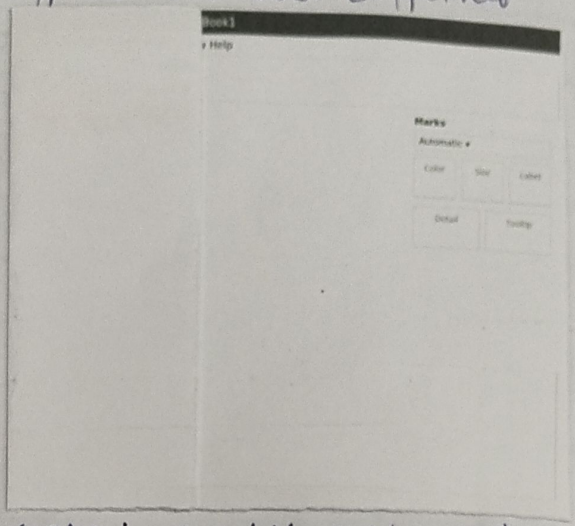
output Postal code's data type is changed from Number to string so it is treated as a category, not a value to sum.

# Practical : 7

Q:7 stepwise procedure with output screenshot

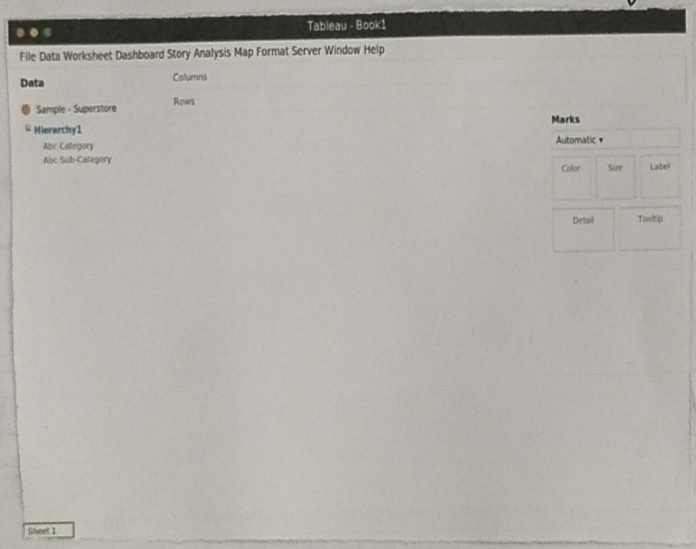
step:1 select the fields to Group

output: category and sub-category are selected together (ctrl+click) in the Data pane.




step:2 create the hierarchy

output: Right-click > hierarchy > create hierarchy puts sub category under category as a new hierarchy field.




step:3 name the hierarchy

output: The hierarchy is named "product hierarchy" and appears at the top of the data pane with an expandable icon.

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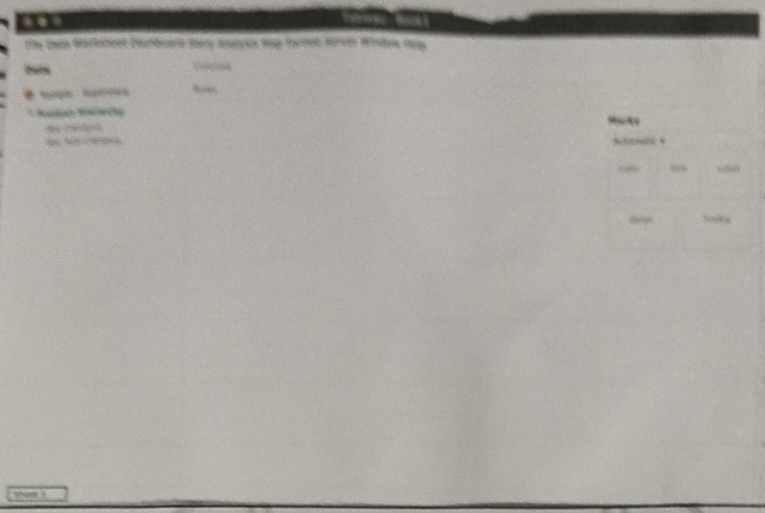
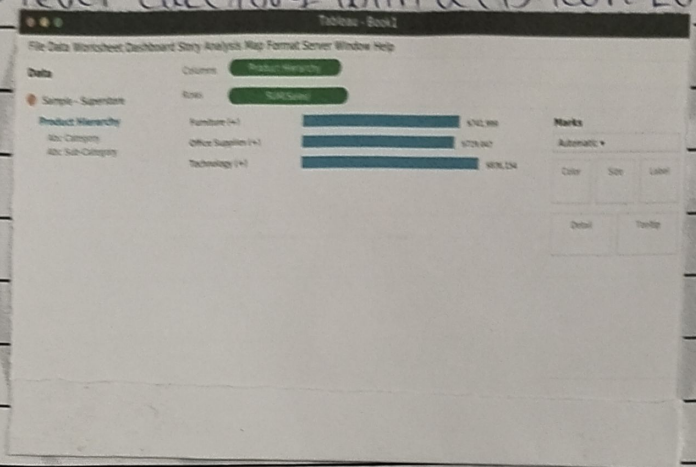


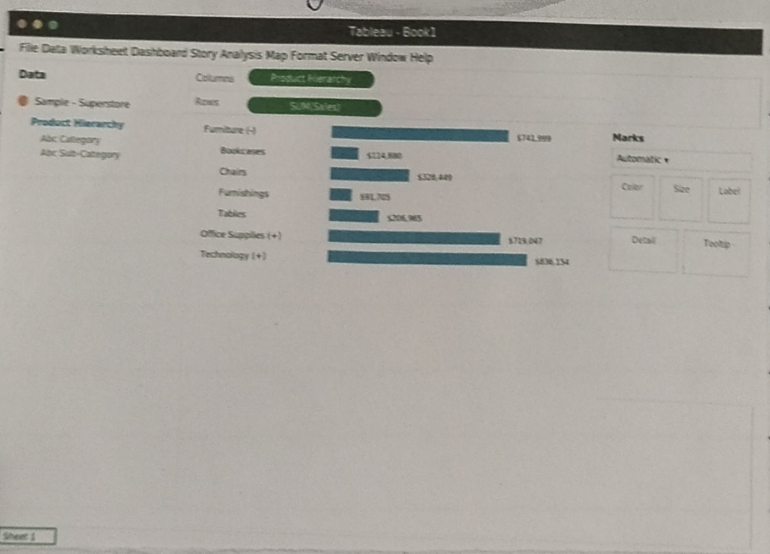
step:4 Drag the Hierarchy to Rows

output: Product hierarchy on Rows initially shows only the top level - Category - with a (+) icon to drill down.



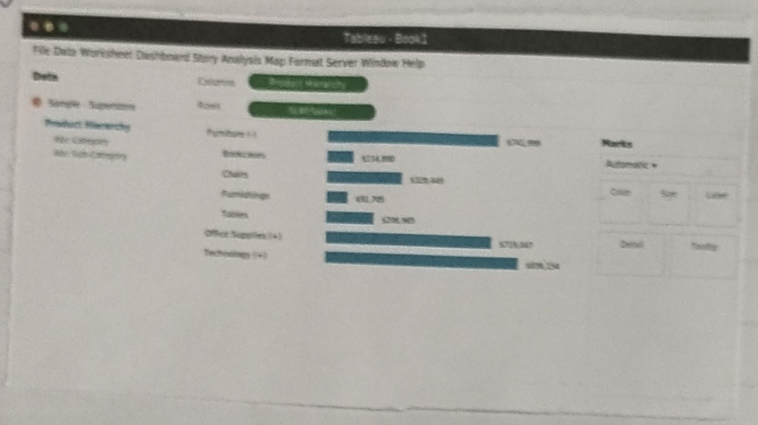

step:5 Expand the hierarchy to Drill Down

output Clicking (+) on Furniture expand it to show Bookcases, chairs, Furnishing and tables with their sales values.



Final output:

The final output is a worksheet where the product hierarchy is expanded for Furniture, revealing its four sub-categories (Bookcase, Chairs, Furnishings, Tables) with individual sales value, while Office supplies and Technology remain collapsed at the Category level.




Conclusion:

Thus, a product hierarchy combining category and sub-category was successfully created in Tableau, allowing users to drill down from category level to sub-category-level sales analysis.

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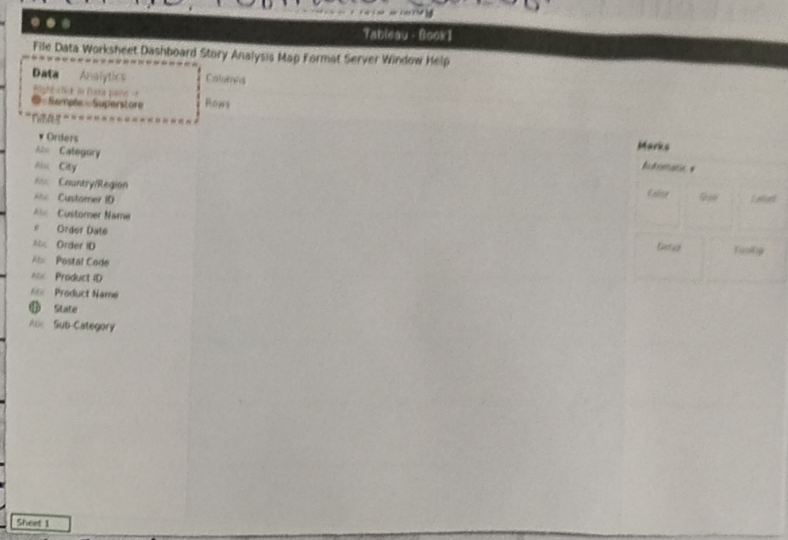
Q:8 stepwise procedure with output screenshots

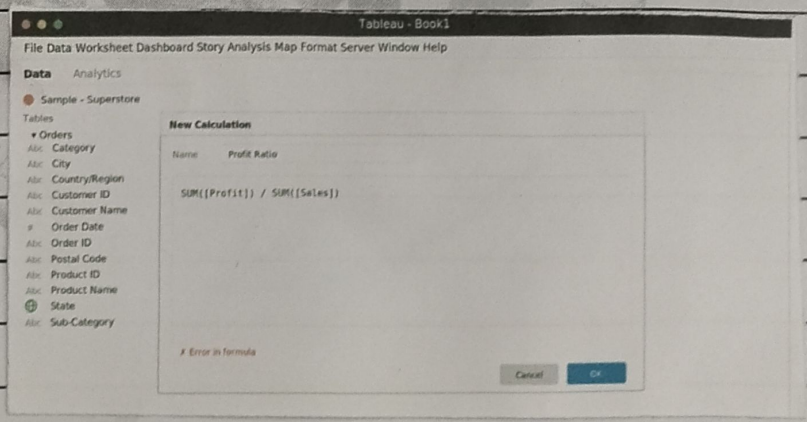
step:1 open the created calculated Field Dialog

output Right-click in the Data pane and choose Create Calculated Field to open the Formula editor.



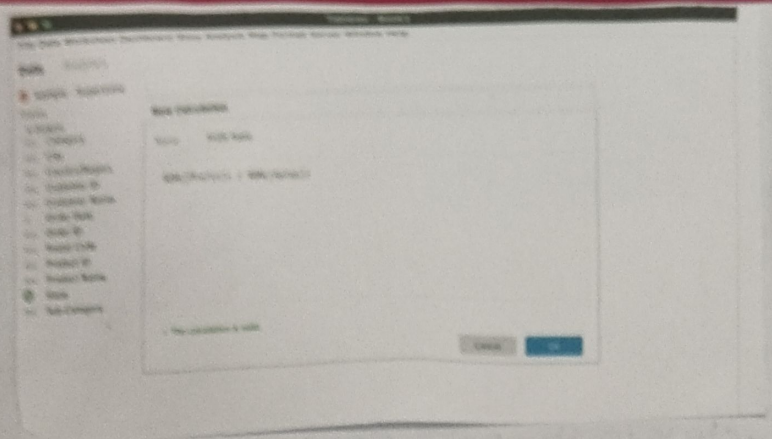

step:2 write the calculatic

output: The formula SUM([Profit])/SUM([Sales]) is entered under the name Profit Ratio.



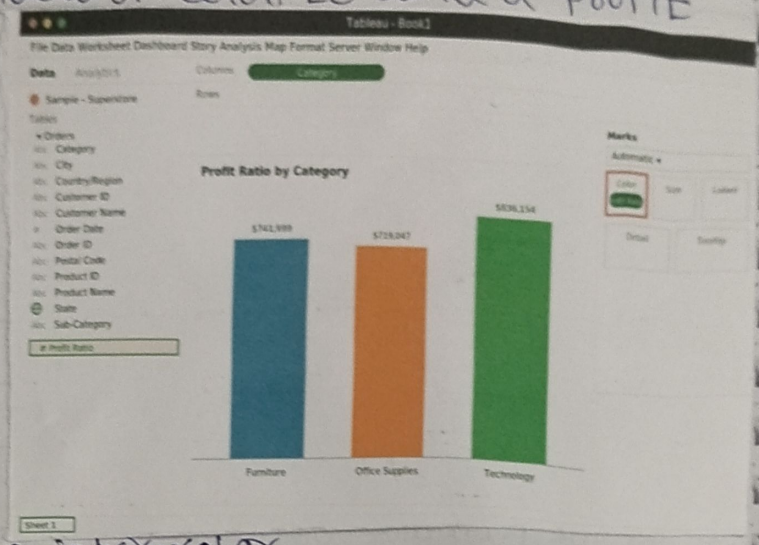

step:3 verify and save the calculated Field

output Tableau confirms "The calculation is valid", clicking OK adds Profit Ratio to the measures lists.



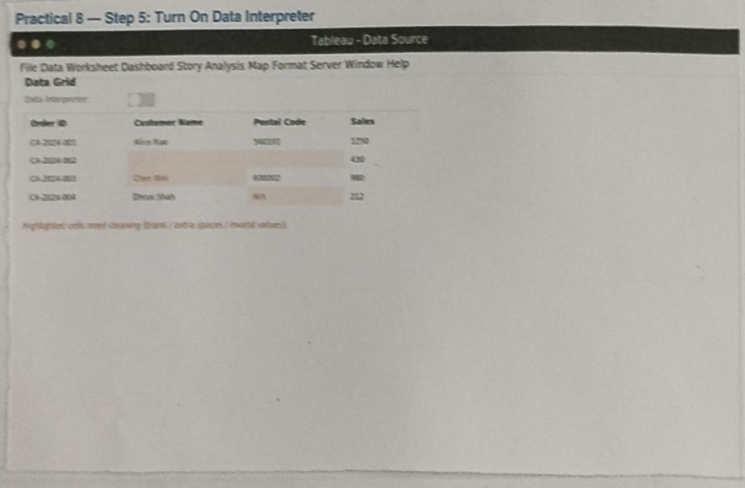
step:4 use the calculated field in a worksheet

output: profit ratio is placed on color to build a "profit ratio by category"




step:5 Turn on the Data interpreter

output: Data interpreter is switched ON; blank, extra-space and invalid values is the grid are higlighted for review.




step:6 Clean Dataset

output: Blank customer name are Filled in, extra spaces Erimmed, and invalid postal codes corrected in the cleaned data grid.

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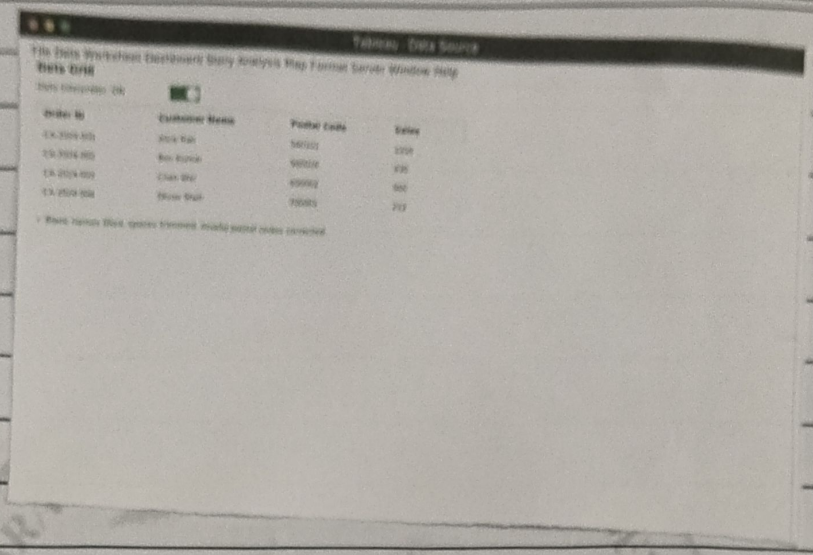
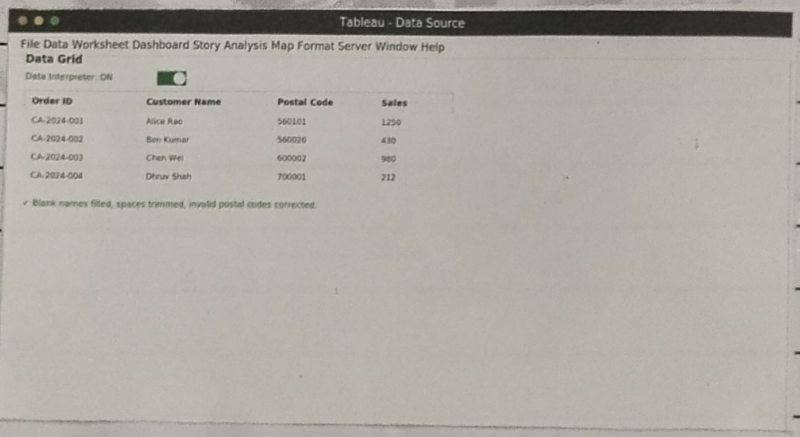



Final output:

The final output shows the cleaned data grid-blank customer name filled, extra whitespace trimmed, and invalid postal codes corrected-alongside the new profit ratio calculated field used earlier to build a profitability chart by category.




Conclusion:

Thus, a calculated field (profit ratio) was created and used in a visualization, and the sample-superstore dataset was cleaned using Tableau's data preparation techniques.

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Unit: 2 SPDT Practical: 1

Q:1 create bar chart and Line charts using Tableau

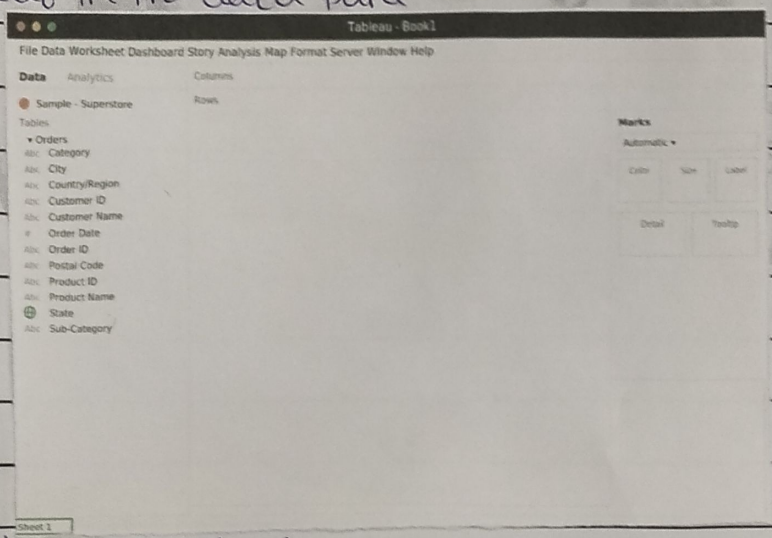
Aim: To create bar-chart and line chart visualization in Tableau using the sample-superstore data sets, comparing sales across categories and analyzing the monthly sales tren

procedure overview:

each step below is shown with the resulting output and a screenshot of the corresponding Tableau screen.

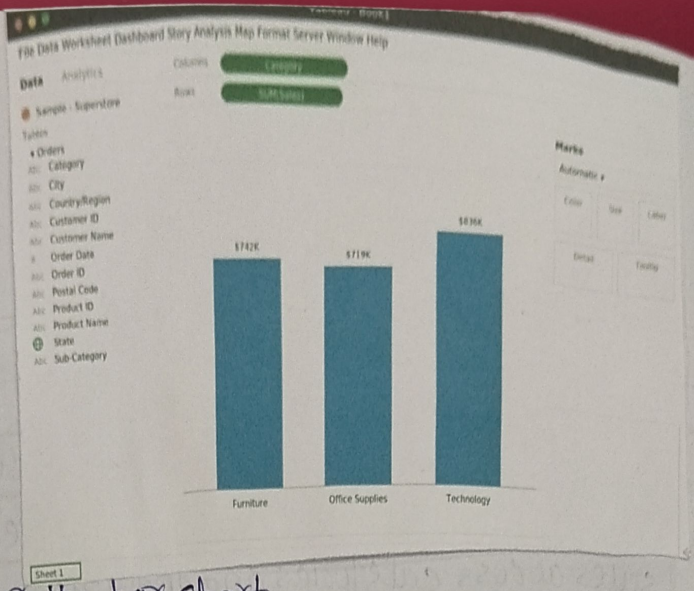
step:1 Connect to the sample-superstore Dataset

output: The dataset connect, orders, people and return tables appear in the data pane.

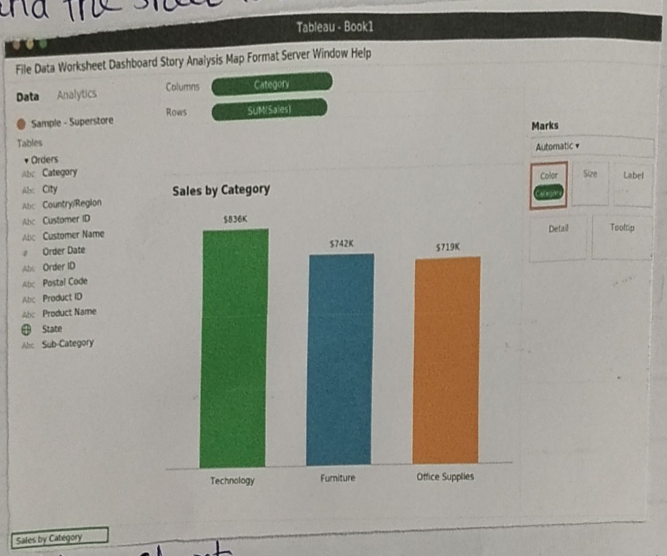



step:2 build a bar chart

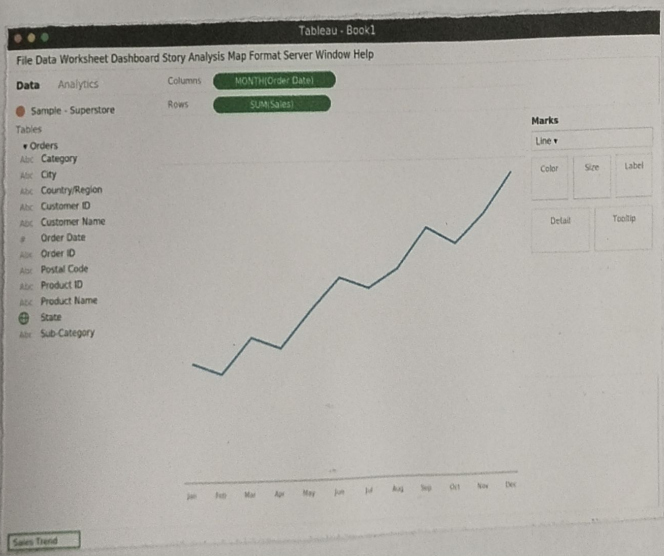
output: category is placed on column and SUM(sales) on Rows; vertical bar chart is Generated automatically.




step:3 sort and colour the bar chart
output: bars are sorted is decending order, coloured by category, and the sheet is titled "sales by category"




step:4 build a Line Chart
output: MONTH(Order Date) is placed on columns, SUM(sales) on Rows, and marks type is change to Line.



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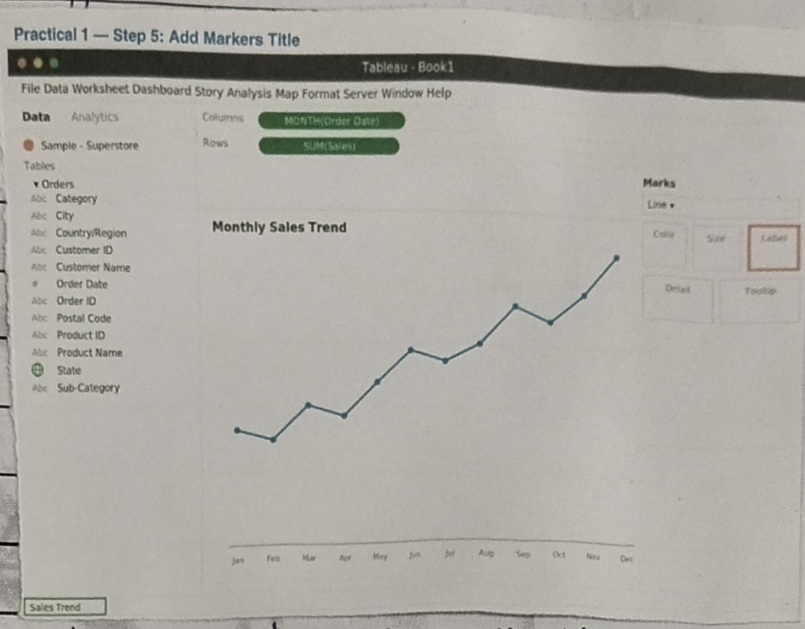
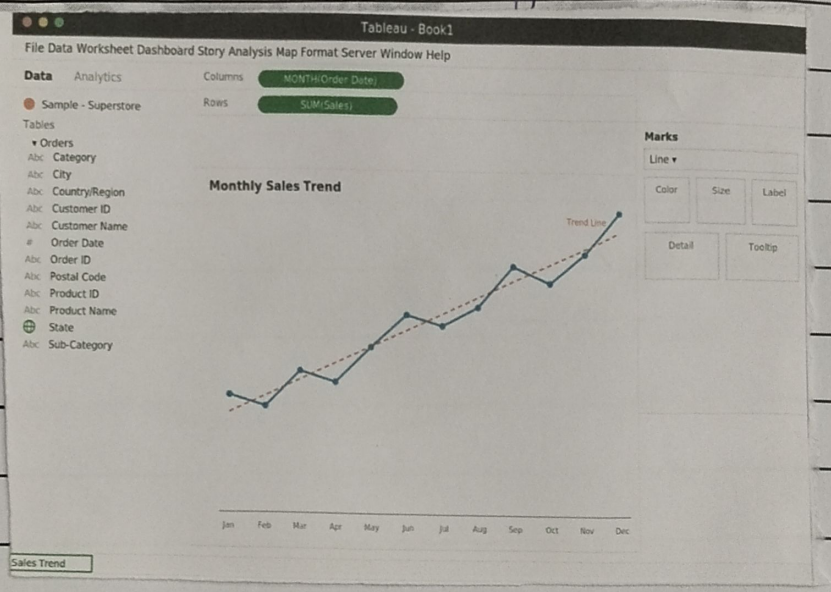
step:5 Add markers and a Title

output: Circular markers appear at each months' data point, the sheet is titled "monthly sales Trends"




step:6 add a Trend Line

output: analysis > Trend Line > show Trend Lines overlays a dashed linear Trend Line confirming the overall sales direction

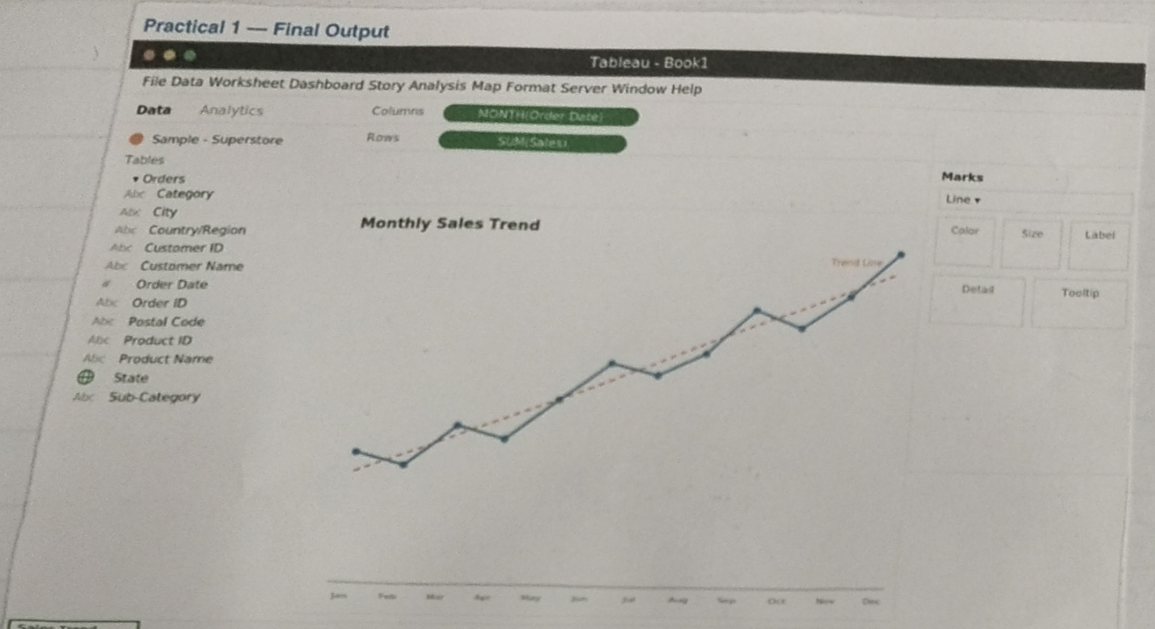



Final output

The Final output a line chart titled "monthly sales Trend", showing SUM(sales) For every month with circular markers at each data point and a dashed red Trend line indicating a clear upward Trend across the year.

## Conclusion:

Thus, bar chart and line chart visualization were successfully created in Tableau to compare category wise-sales and analyse the monthly sales trend with trend line.



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Spdt Practical: 2

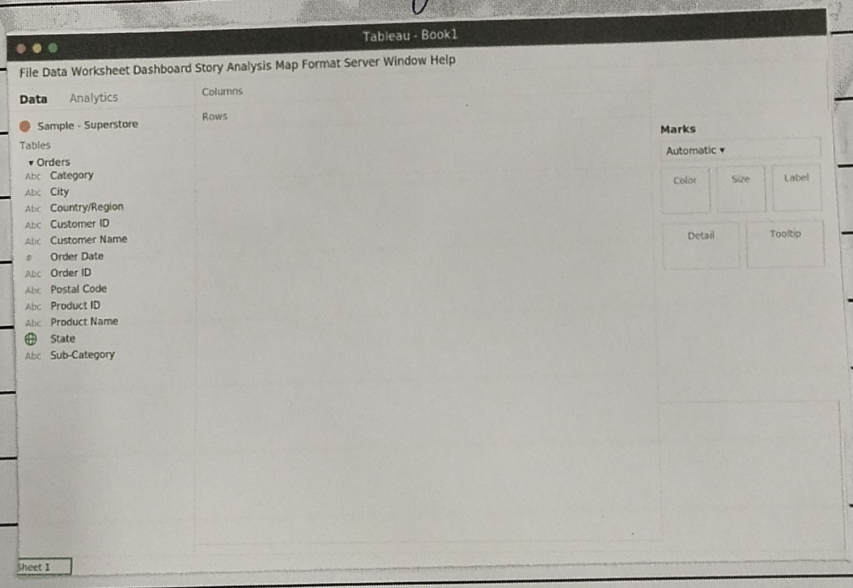
Q:2 create Heatmaps and highlight Tables For Data visualization.

Aim: To create a heat map and a highlight Table in Tableau to visualize sales across category and Region From the sample superstore dataset.

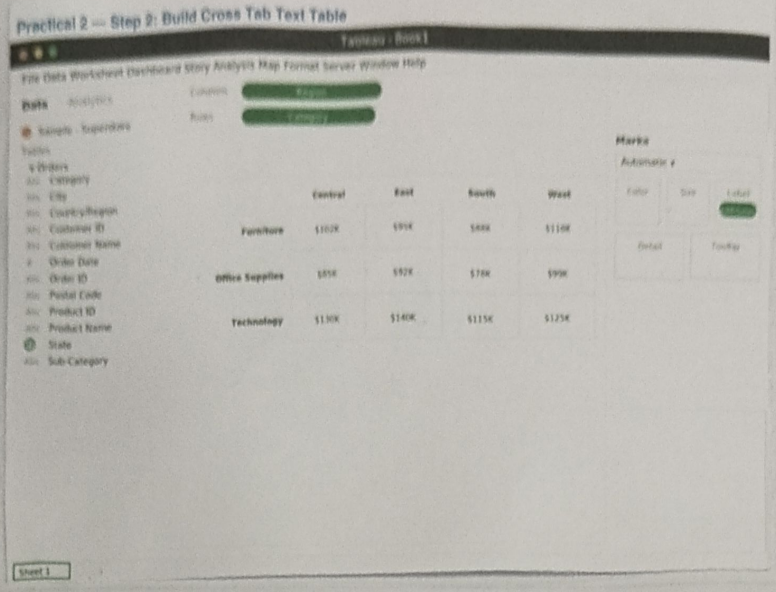
procedure overview
each step below is shown with the resulting output and a screenshots of the corresponding Tableau screen

step: 1 Connect to Data & open a worksheet

output: The sample-superstore data connects and a blank worksheet open, ready For the visualization.

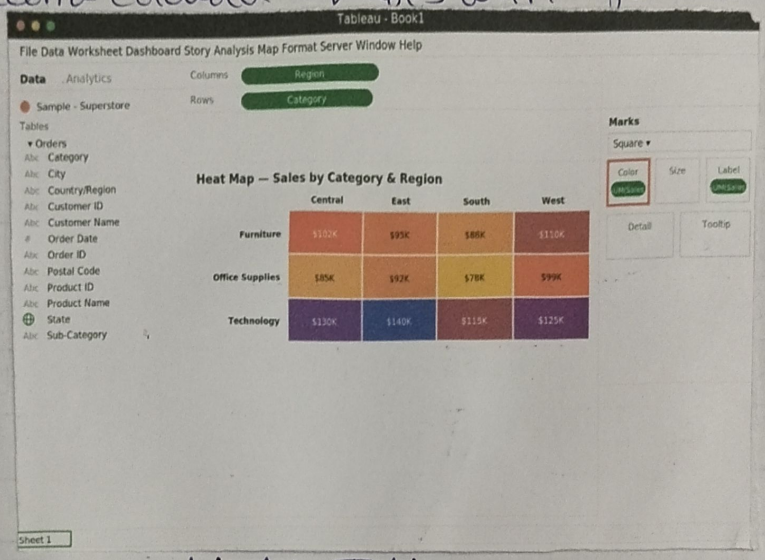



build a Cross-tab (Text table)
Region on Columns, category on Rows and sales on text produce a plain text table of sales values.



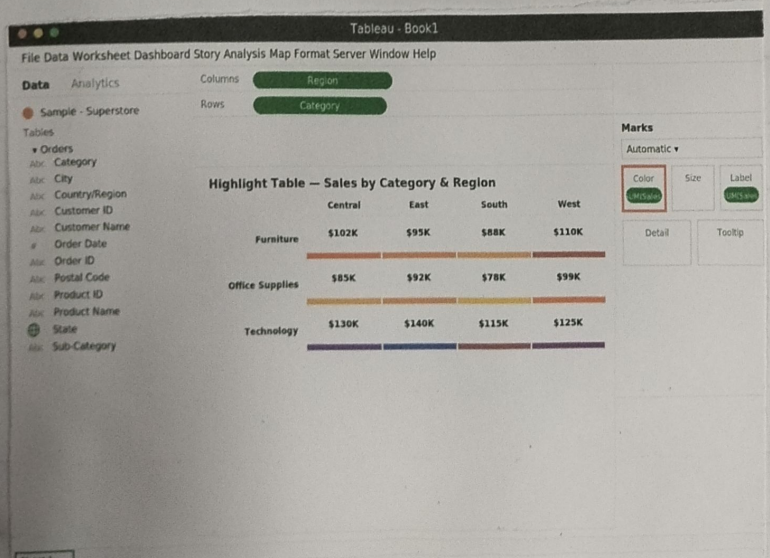

step:3 Convert to heat Map

output: mark type is changed to square and sales is placed on color. Cells become coloured squares with a gradient legend.




step:4 Convert to a highlight Table

output: marks are kept as text with sales colors cell background are shaded while exact value stays readable.



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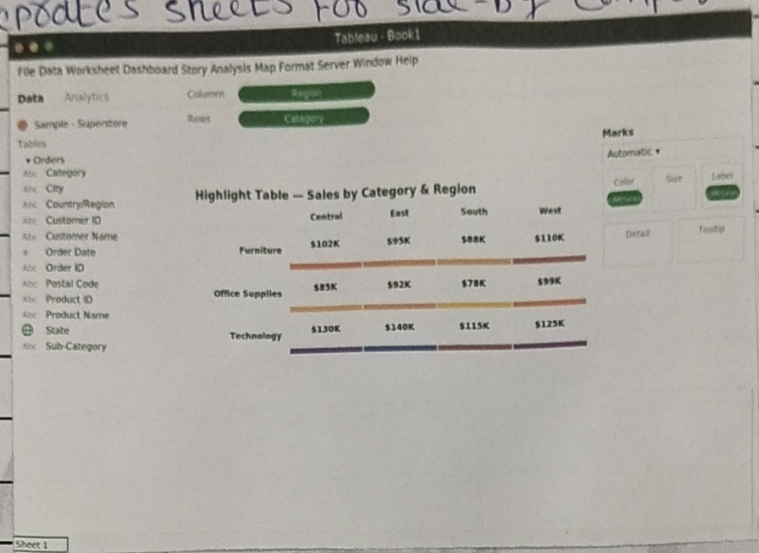
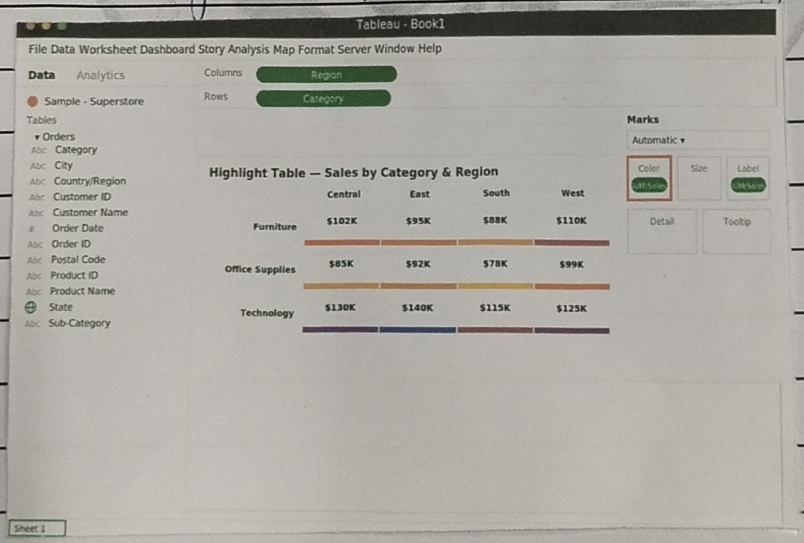
step:5 Compare and Title both views

output: The heat map and highlight table and titled and saved as seprates sheets for side-by comparison.




Final output:

The final output highlight table title "highlight table - sales by category & region", where every category-region cell shows the exact sales figure as bold text with a coloured strip beneath it indicating relative magnitude - combining readability with quick visual scanning.




Conclusion: Thus, a heat map and a highlight Table were successfully created in Tableau to visualize sales across category and region making high-low performing combinations easy to spot at glance.

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# SPDT Practical: 3

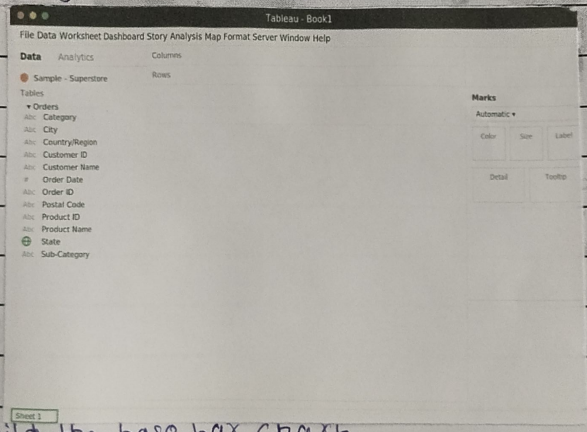
Q:3 create bullet charts for comparative Analysis

Aim: To create a bullet chart in tableau comparing actual sales against Target sales for each region.

procedure overview:
each step below is shown with the resulting output and a screenshot of the corresponding tableau screen

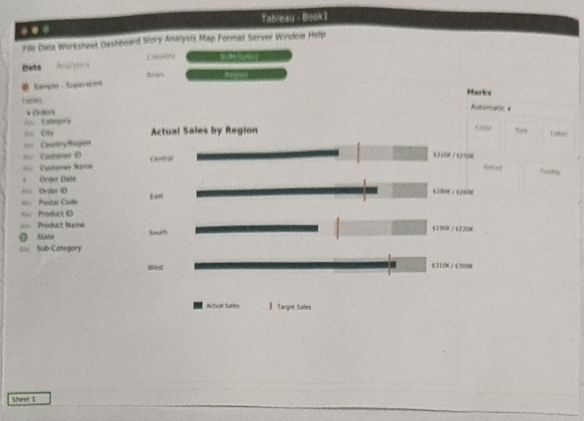
step: 1 Connect to Data with actual & Target measures

output: sample - superstore (sales) and sales Target extract (Target sales) are connected and blended on Region.



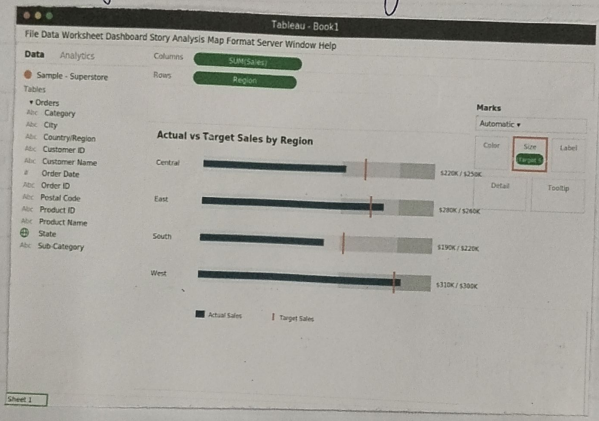

step: 2 build the base bar chart

output Region is placed on Rows and SUM(sales) on columns, producing a horizontal bar of actual sales by Region.



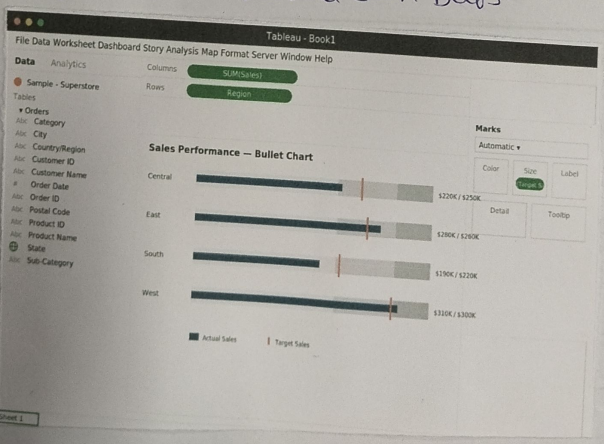

step:3 Add the target measures

output: SUM(Target sales) is added via size & reference tick line for the target appears on every bar.




step:4 Add Qualitative Background bands

Output: Three shaded bands (light/medium/Grey) representing performance ranges appears behind each bars.



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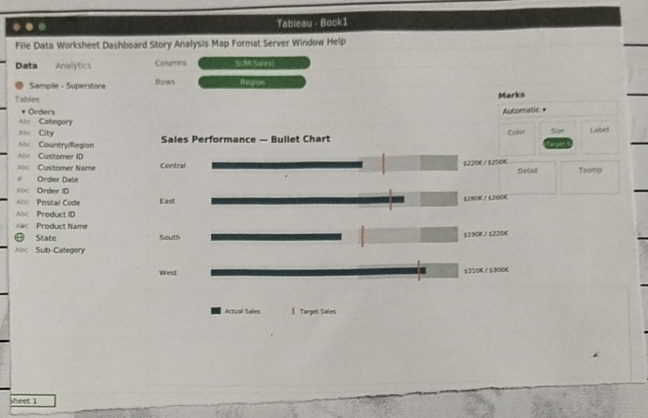
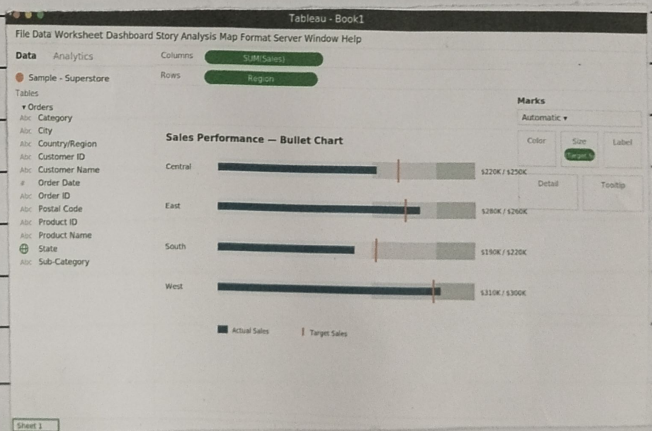
Step:5 Format and Title the bullet chart

output: Bars, target ticks and bands are labelled and titled "sales performance - Bullet chart" with legend.




Final output:

The Final output is Bullet chart titled "sales performance bullet chart", showing a dark-blue actual sales bar and a red target sales tick for central, east, south and west against Grey qualitative performance bands- east and west are seen exceeding their target their while central and south fall short.



Conclusion:

Thus, a bullet chart was successfully created in Tableau to compare actual sales against Target sales for each region, using qualitative background bands for quick performance assessment.

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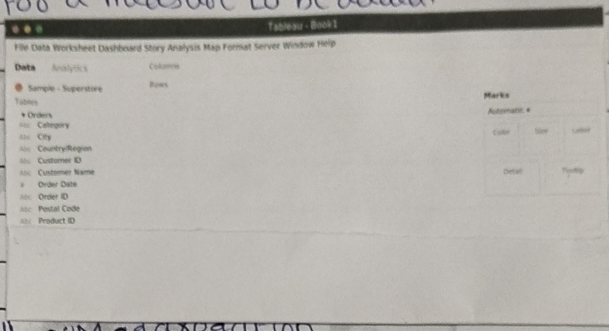
# Spdt Practical: 4

Q:4 Connect to Data & builds the base view

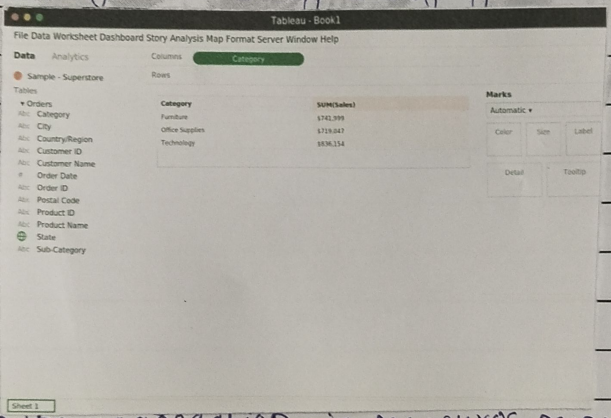
step:1 Connect to Data & build the base view

output: category is placed on columns, with the rows shelf, empty, ready for a measure to be added.



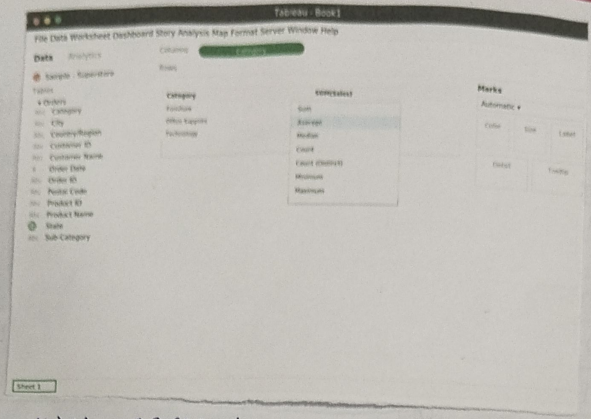

step:2 apply the Default SUM aggregation

output: sales is dragged to rows; SUM(sales) is computed and displayed for every category.

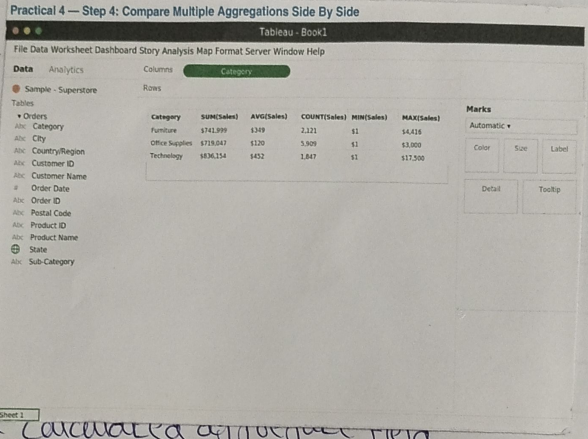



ep:3 change the aggregation via measures menu

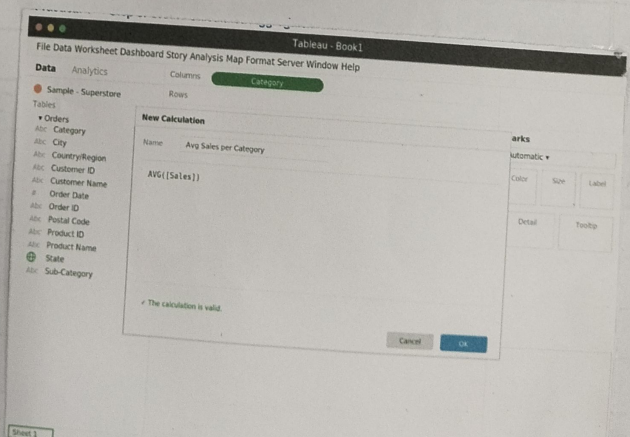
utput: right-click sum(sales) > measure opens a list of aggregation (sum, average, median, count, min, max); average is selected.




step:4 Compare multiple aggregation side by side
output: a table shows SUM, AVG, COUNT, MIN and MAX of sales for every category in one view.




step:5 Create a calculated field
output: a calculated field, AVG([Sales]), is created and validated for reuse across the workbook.



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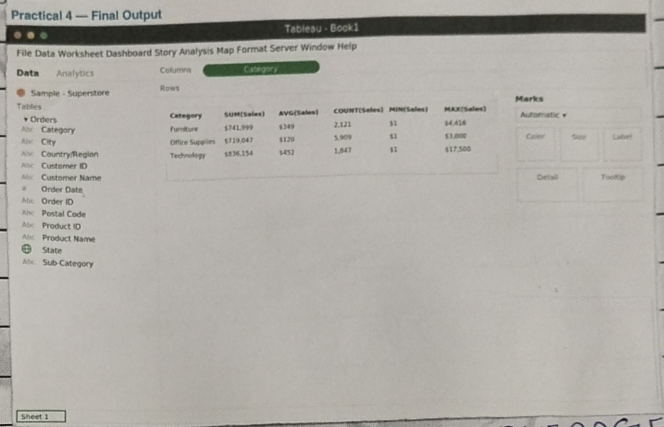
Final output:

The final output is a comparison table listing SUM, AVG, COUNT, MIN and MAX of sales for Furniture, office suppliese and Technology, clearly showing show how each aggregate function summarize the same underlying data differently




Conclusion:

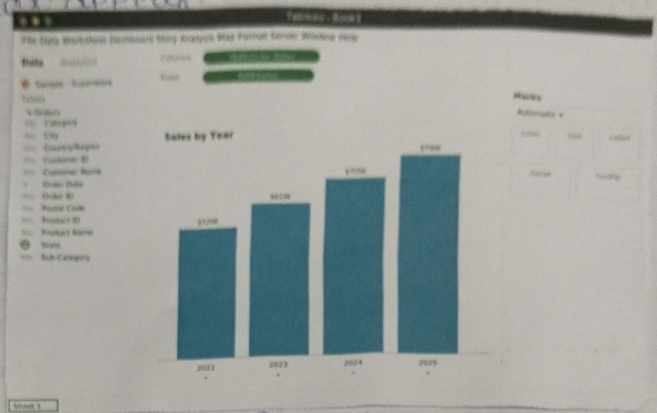
Thus, different aggregate function - SUM, AVERAGE, COUNT, MIN and MAX - were successfully applied to the sales measures in Tableau and Compared across product categories.

# Practical: 5

Q:5 step wise procedure with output screenshot

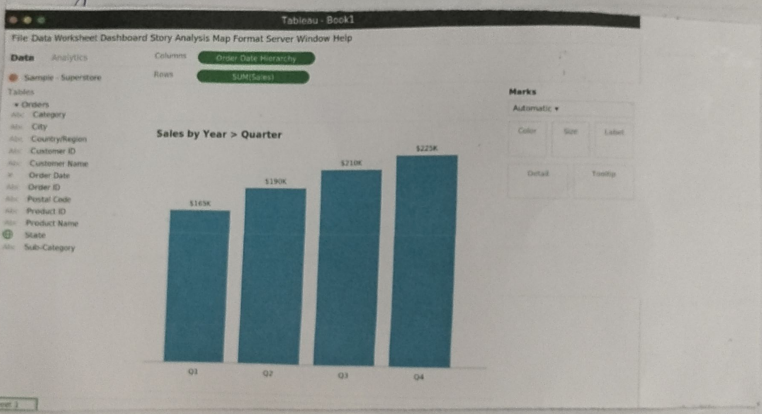
Step: 1 Drag order Date to columns

Output: YEAR (order Date) is auto-Generated on columns. Bar Chart of sales by year appear.




step: 2 Drill Down to the Quarter Level

output: expanding the date hierarchy break year down into sales by quarter




step: 3 Create a calculated field using DATENAME

output: Order moth Name = DATENAME('month', [Order Date]) is created and validated.

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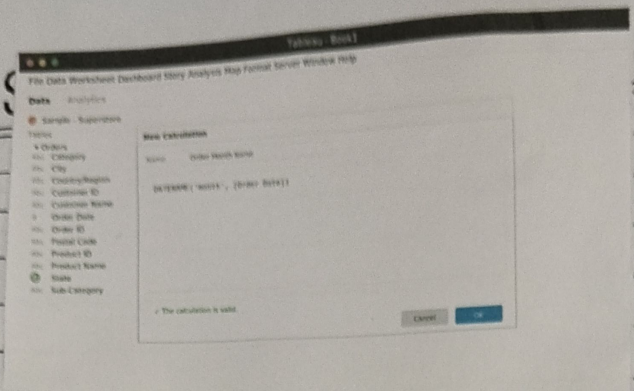
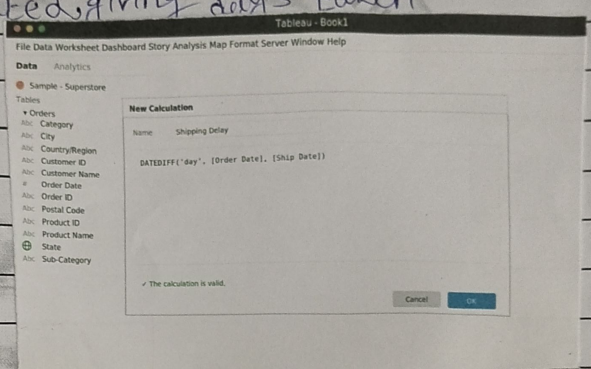


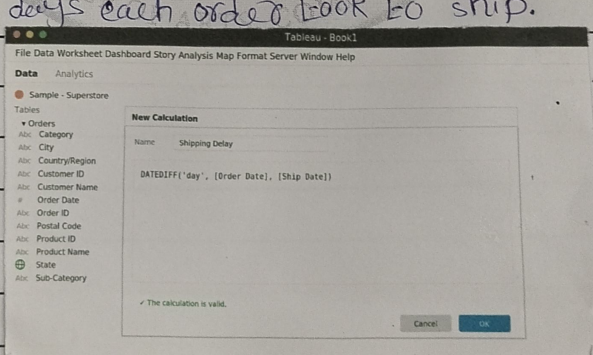
step:4 create a calculated field using DATEDIFF

output: shipping Daly = DATEDIFF('day', [Order Date], [ship Date]) is created, giving days taken




Final Output

The Final output is validated "shipping day" calculated field using DATEDIFF('day', [order Date], [ship Date]), which can now be used anywhere in the workbook to analyse how many days each order took to ship.




Conclusion:

Thus, Tableau ... and date functions (DATENAME, DATEDIFF) were successfully used to transform and analyse order date ship date field in the sample - superstore dataset.

# Practical: 6

Q: 6 apply text and logical functions for data manipulation

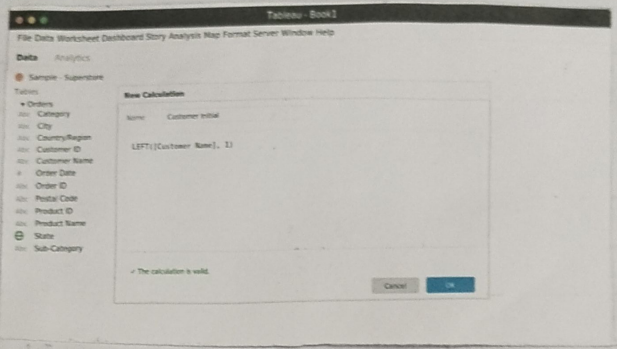
Aim: To use text function (LEFT) and logical function (IF/THEN/ELSE) to create calculated field tableau and apply them to classify and visualize sample-superstore

## Procedure overview

Each step below is show with resulting output and screenshot corresponding Tableau screen.

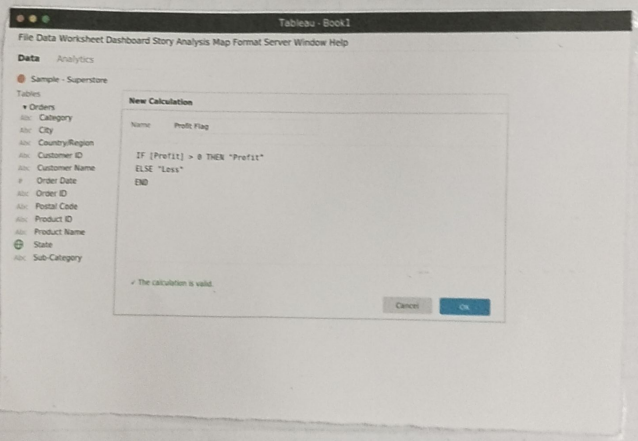
Step: 1 Create a text function calculated field

Output: customer initial = LEFT([Customer Name], 1) is created and validated




Step: 2 Create a Logical function calculated field

Output: profit Flag = IF [PROFIT] > 0 THEN "profit" "Loss" END is created and validated.



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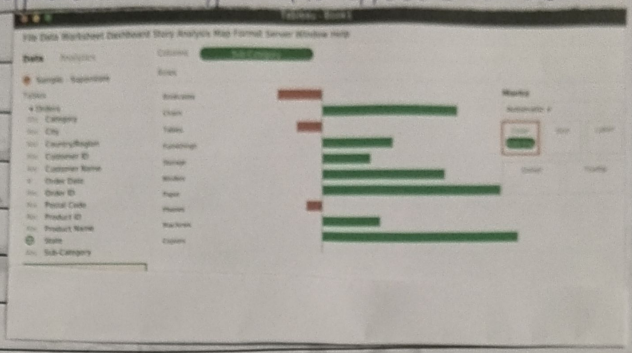
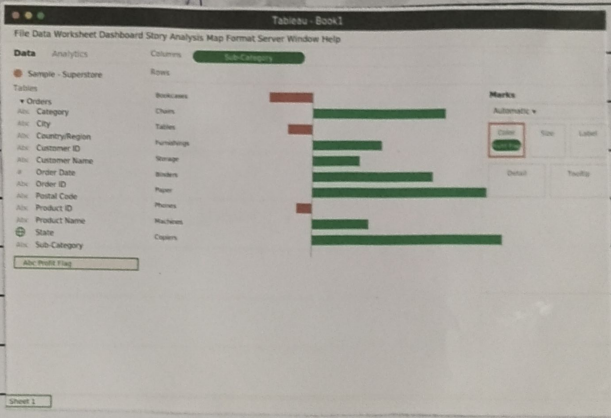
step: 3 Use the calculated field in a worksheet

output: Profit Flag is placed on color to build a diverging bar chart of sub-categories, showing profit (green) vs Loss (red)



step 4 Final output:

The final output is diverging bar chart - sub categories coloured by profit flag calculated field - sub categoriese such as bookcases, Tables and phones appear in red (Loss) while the rest (appear) in green (profit), making unprofitable product line immediately visible.




Conclusion:

Thus, text and logical calculated field (LEFT, IF / THEN / ELSE) were successfully created in Tableau and used to classify and visualize profit and Loss across sub-categoriese.

# Practical :7

Q:7 Implement Table Calculation and Level of detail (LOD) expression

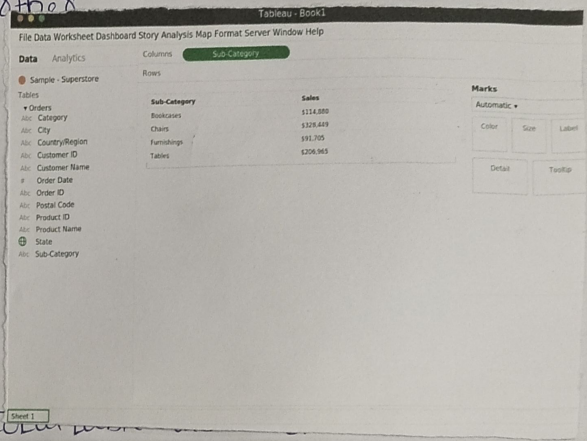
Aim: To apply Table calculation (Running, Total, Rank) and a Level of Detail (LOD) expression in Tableau to analyze Sales by sub-category within sample-superstore dataset

Procedure overview:

Each step below is shown with resulting output and screenshot of the corresponding Tableau screen

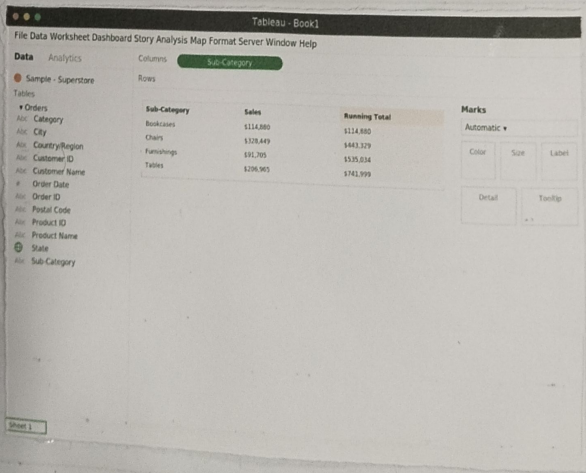
Step: 1 build the base view

Output: sub category is placed view with SUM (sales), Forming the base table For further




Step: 2 Add a Running Total

Output: Right-click SUM(sales) > Quick Table calculation > Running Total adds a Columns that accumulates sales progressively.



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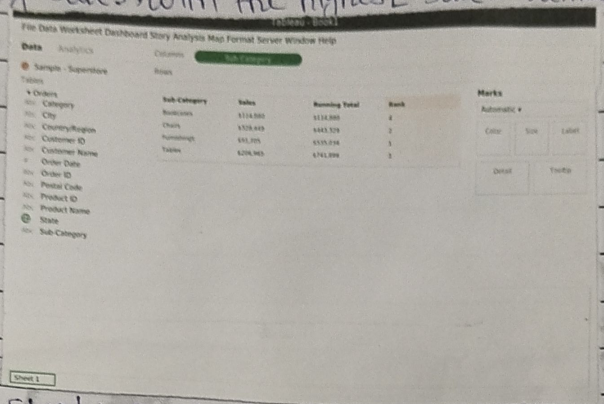
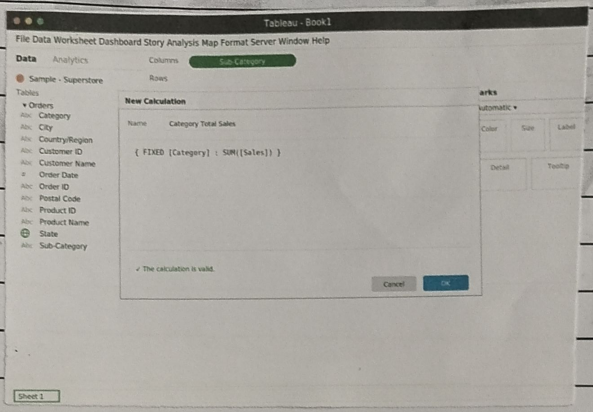
step:3 Add a Rank Table calculation

output a Rank column show each sub-category's position by sales, with the highest sales ranked.



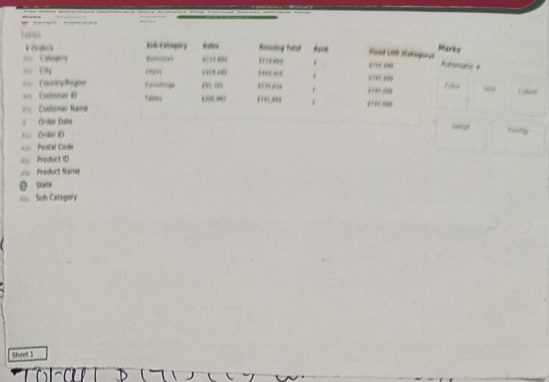

step:4 C

output a Fixed LOD calculated field, {FIXED [Category] : SUM ([Sales])}, is created and validated.



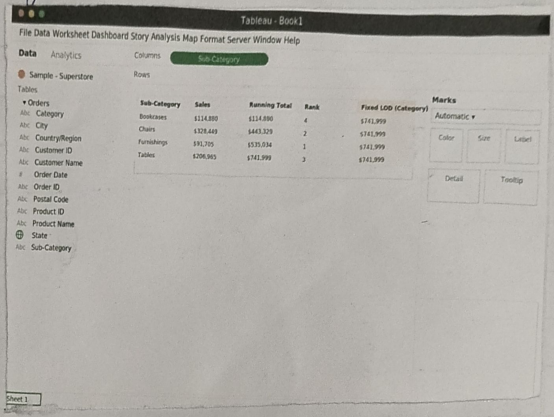

step:5 Compare Table calculation vs LOD Result

output a Final table show sales, Running Total and Rank alongside the Fixed LOD category Total, highlighting how each recalculates differently.



Final output:

"The Final output is
under Furniture, s
Running Total and
Fixed LOD category total
Constant for every row regardless of the view's structu




Conclusion:-

Thus, table calculation (Running, Total, Rank) and a Fixed LOD expression were successfully applied in tableau to analyse sales by sub-category, demonstrating the difference between view dependent and level independent calculations.